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Common rail cooling plate

Abstract

A cooling plate assembly to facilitate the cooling of electronic components, the cooling plate assembly including a cooling plate and a cooling chamber. The cooling chamber comprising a surface configured to receive a component to be cooled, a cooling chamber inlet, a common rail, a collection cavity, a plurality of cooling units fluidly coupled between the common rail and collection cavity, and a cooling chamber outlet fluidly coupled to the collection cavity.

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Background/Summary

FIELD

(1) The present disclosure relates generally to cooling plates for cooling electronic components, such as IGBT modules.

BACKGROUND

(2) During the use of electronic systems, electronic components may release heat that can damage

and affect the operationality of the system. Therefore, it is necessary to be able to dissipate the heat created by electronic components in an efficient manner. Known cooling systems may provide uneven cooling, allowing some components to be cooled while others continue to release heat into the electronic system. Uneven flow splits may cause uneven cooling since different pathways will receive different amounts of coolant. Furthermore, electronic components positioned closer to the coolant inlet may be exposed to lower temperature coolant than the electronic components positioned at the end of a cooling pathway due to the coolant absorbing the heat as it flows through the cooling pathway.

(3) Referring initially to FIG. **1**, a known cooling device **10** is shown. Cooling device **10** comprises a cooling plate **20** and cooling chamber **22**. Cooling plate **20** includes a first surface **12** and a second surface **14**. Cooling chamber **22** is adjacent to first surface **12**. A plurality of electrical components or other heat-producing objects (not shown) are coupled to second surface **14**, which is opposite of first surface **12**. Cooling chamber **22** includes two mirrored cooling loops **30** and a coolant outflow cavity **60**. The cooling loops **30** each comprise a cooling loop inlet **40**, a passageway **42** defined by passageway, and a cooling loop outlet **50** that is fluidly coupled to the coolant outflow cavity **60**.

(4) Cooling loops **30** are continuous loops that run from loop inlet **40** to loop outlet **50**. Coolant flows into loop inlet **40**, through passageway **42** enclosed by passageway walls **44**, and out loop outlet **50**. While flowing through passageway **42**, the coolant absorbs heat through first surface **12** from the electrical components coupled to second surface **14**, thereby cooling the electrical components. Once the coolant flows out of loop outlet **50**, the coolant is collected in coolant outflow cavity **60** before flowing out of cavity **60**.

SUMMARY

(5) The present disclosure provides a cooling device. In an embodiment, the cooling device comprises: a surface configured to receive a component to be cooled; a cooling chamber inlet; a common rail fluidly coupled to the cooling chamber inlet; a collection cavity; a plurality of cooling units, each cooling unit located adjacent to the surface and comprising a passageway fluidly coupled between the common rail and the collection cavity; and a cooling chamber outlet fluidly coupled to the collection cavity. In some embodiments, the common rail is a central rail, and the plurality of cooling units are located on opposite sides of the central rail. In another embodiment, the common rail, plurality of cooling units and collection cavity are configured such that a pressure of a coolant fluid within the common rail is higher than a pressure of a coolant fluid within the collection cavity, and the coolant fluid flows into each of the plurality of cooling units at a generally equal velocity.

(6) In various embodiments, the passageway is serpentine, optionally defined by adjacent linear segments connected by turns. In certain embodiments, each cooling unit further comprising a plurality of posts spaced apart from one another along a length of the passageway.

(7) According to further embodiments, a common rail cooling device is provided. The common rail cooling device comprises: a first cooling plate, the first cooling plate comprising a first surface, and a second surface configured to receive one or more electronic components; a cooling chamber on the first surface of the cooling plate, the cooling chamber configured to receive cooling fluid and comprising: a common rail chamber; a collection cavity; and a plurality of cooling units including passageways fluidly coupled between the common rail and the collection cavity. In some embodiments, the common rail cooling device further comprises a second cooling plate opposite the cooling chamber from the first cooling plate, the second cooling plate including a first surface adjacent the cooling chamber and a second surface configured to receive one or more electronic components.

(8) In certain embodiments, each of the one or more electronic components is located adjacent to a plurality of the cooling units. In further embodiments, each of the plurality of cooling units comprises a serpentine passageway. In other embodiments, the passageway of each cooling unit

includes a plurality of segments, each of the plurality of segments coupled to the next segment in the plurality of segments by a turn.
(9) In various embodiments, the plurality of cooling units are connected in a parallel fluid circuit between the common rail chamber and the collection cavity. In some embodiments, the common rail, plurality of cooling units and collection cavity are configured such that a pressure of a coolant fluid within the common rail is higher than a pressure of a coolant fluid within the collection cavity, and the coolant fluid flows into each of the plurality of cooling units at a generally equal velocity.
(10) According to other embodiments, a cooling assembly is provided. The cooling assembly comprises a first member comprising: a first cooling plate, the first cooling plate including a first surface, and a second surface configured to receive one or more electronic component to be cooled; and a plurality of wall segments extending from the first surface, the wall segments defining a common rail cavity portion, a collection cavity portion, and a plurality of cooling unit passageway portions between the common rail cavity portion and the collection cavity portion; and a second member coupled to the first member, comprising: a second cooling plate, the second cooling plate including a first surface, and a second surface configured to receive one or more electronic components to be cooled; and a plurality of wall segments extending from the first surface, the wall segments defining a common rail cavity portion, a collection cavity portion, and a plurality of cooling unit passageway portions between the common rail cavity portion and the collection cavity portion; wherein when the first member and the second member are coupled, the wall segments of the first member are coupled to the wall segments of the second member to define a common rail cavity, a collection cavity, and a plurality of cooling unit passageways between the common rail cavity and the collection cavity.
(11) According to further embodiments, the first member and the second member each comprise: wall segments defining a cooling chamber inlet portion coupled to the common rail portion; and wall segments defining a cooling chamber outlet portion coupled to the collection cavity portion; wherein when the first member and the second member are coupled, the wall segments define a cooling chamber inlet and a cooling chamber outlet. In certain embodiments, the plurality of cooling unit passageway portions of the first member and the plurality of cooling unit passageway portions of the second member each are configured such that a pressure of a coolant fluid within the common rail is higher than a pressure of a coolant fluid within the collection cavity, and the coolant fluid flows into each of the plurality of cooling units at a generally equal velocity.
(12) In certain embodiments, the plurality of wall segments of the first member and the plurality of wall segments of the second member each define a serpentine plurality of cooling unit passageway portions connected by turns. In another embodiment, the plurality of cooling unit passageway portions of the first member and the plurality of cooling unit passageway portions of the second member each comprise a plurality of posts spaced apart from one another along a length of the plurality of cooling unit passageway portions.
(13) In various embodiments, the first member and the second member are coupled, the plurality of posts of the first member and the plurality of posts of the second member are spaced apart by a distance. In another embodiment, the one or more electronic component to be cooled on the first cooling plate is located adjacent to the plurality of cooling unit passageway portions of the first cooling plate, and the one or more electronic component to be cooled on the second cooling plate is located adjacent to the plurality of cooling unit passageway portions of the second cooling plate. In certain embodiments, the plurality of cooling unit passageway portions of the first cooling plate is connected in a parallel fluid circuit between the common rail portion and the collection cavity portion, and the plurality of cooling unit passageway portions of the second cooling plate is connected in a parallel fluid circuit between the common rail portion and the collection cavity portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above-mentioned and other features and advantages of this disclosure, and the manner of obtaining them, will become more apparent, and will be better understood by reference to the following description of the exemplary embodiments taken in conjunction with the accompanying drawings, wherein:

(2) FIG. **1** is a top view of a known cooling assembly coupled to a cooling plate;

(3) FIG. **2** is a perspective view of two cooling assemblies including two cooling plates, and IGBT modules coupled to the cooling plates;

(4) FIG. **3** is a cross-sectional side view of the cooling assemblies and corresponding cooling plates of FIG. **2**;

(5) FIG. **4** is a cross-sectional view of one cooling assembly of FIG. **2**, showing an interior first surface of the cooling plate;

(6) FIG. **5** is an exemplary image of fluid pressure within the cooling assembly of FIG. **4** during operation;

(7) FIG. **6** is an exemplary image of fluid velocity within the cooling assembly of FIG. **4** during operation; and

(8) FIG. **7** illustrates a thermal comparison between the cooling assemblies of the present disclosure and the cooling assembly of FIG. **1**.

(9) Corresponding reference characters indicate corresponding parts throughout the several views. Although the drawings represent embodiments of various features and components according to the present disclosure, the drawings are not necessarily to scale and certain features may be exaggerated in order to better illustrate and explain the present disclosure. The exemplification set out herein illustrates an embodiment of the invention, and such an exemplification is not to be construed as limiting the scope of the invention in any manner.

DETAILED DESCRIPTION

(10) For the purposes of promoting an understanding of the principles of the present disclosure, reference is now made to the embodiments illustrated in the drawings, which are described below. The exemplary embodiments disclosed herein are not intended to be exhaustive or to limit the disclosure to the precise form disclosed in the following detailed description. Rather, these exemplary embodiments were chosen and described so that others skilled in the art may utilize their teachings.

(11) To dissipate unwanted heat from an electronic system, cooling plates may be used. Cooling plates can have a coolant or other heat absorbing material that is run through a cooling assembly that is in close proximity to heat emitting components of the electronic system. The coolant absorbs the heat from components such that the electronic system is cooled. The configuration of the cooling assembly and cooling plates may affect the efficacy of the cooling.

(12) As seen in FIGS. **2-4**, a cooling assembly **100** of the present disclosure is illustrated. The cooling assembly **100** may comprise two cooling devices **110***a*, **110***b*. Each cooling device **110** may comprise a plate **116***a*, **116***b* and a cooling chamber **120***a*, **120***b*. First cooling plate **116***a* has a first surface **112***a* (FIG. **3**) and a second surface **114***a*. Similarly, second cooling plate **116***b* has a first surface **112***b* and a second surface **114***b*. Cooling assembly **100** may be used to cool a plurality of electronic devices by coupling electronic devices to cooling devices **110***a* and **110***b*. Cooling devices **110***a* and **110***b* may also be used individually to cool electronic devices. To cool electronic devices **115** using cooling assembly **100**, first cooling device **110***a* and second cooling device **110***b* may be positioned such that first surface **112***a* of first cooling plate **116***a* faces first surface **112***b* of second cooling plate **116***b*. As shown in FIG. **3**, passageway walls **137** of cooling units **130** of cooling chambers **120** abut each other such that cooling chambers **120***a* and **120***b* share their

respective voids and passageways **136** as described further herein.
(13) A plurality of electronic components **115** may be removably coupled to the one or both of the second surfaces **114***a*, **114***b*. Electronic components **115** may include components such as IGBT power modules. In other embodiments, rather than electronic components, another heat source may be coupled to second surface **114**. Each electronic component **115** may be equally spaced apart across second surface **114***a* and/or second surface **114***b*, unequally spaced apart across second surface **114***a* and/or second surface **114***b*, or positioned immediately adjacent to one another. Each electronic component **115** may correspond with a cooling unit **130** (FIG. **4**) as described further herein.
(14) While FIG. **2** illustrates four electrical components **115** coupled to each cooling plate **116***a*, **116***b*, other embodiments may include a greater or smaller number of electrical components. For example, other embodiments may include one, two, three, five, six, seven, eight, nine, or ten electrical components. In yet other embodiments, more than ten electrical components may be included. To increase or decrease the number of electrical components **115**, each of the cooling plates **116** are increased or decreased in length and a greater or lesser number of corresponding cooling units **130** (FIG. **4**) are included so that each electrical component corresponds to a cooling unit **130** (FIG. **4**) as described further herein.
(15) Referring to FIG. **4**, cooling device **110***a* is illustrated. Cooling chamber **120***a* may be adjacent to first face **112***a* of cooling plate **116***a*. Because cooling chamber **120***b* may be substantially similar to cooling chamber **120***a*, descriptions of cooling chamber **120***a* may also describe cooling chamber **120***b*. Cooling chamber **120***a* may comprise a cooling chamber inlet **122**, a common rail **124**, a plurality of cooling units **130**, and a collection cavity **160**.
(16) Axis X divides cooling chamber **120***a* into two parts down the length of cooling chamber **120***a* such that axis X runs through cooling inlet **122** and collection cavity **160**. The cooling chamber **120***a* may be generally mirrored across axis X. Primary input port **118** may be fluidly coupled to a cooling chamber inlet **122** to allow a coolant fluid to flow into cooling chamber **120**. Cooling chamber inlet **122** may supply the coolant fluid from primary input port **118** to the cooling chamber **120***a*. Cooling inlet **122** is sized and shaped to have a width that is larger than outlets **150** of the cooling units, discussed below. Once coolant passes through cooling chamber inlet **122**, the coolant enters a common rail **124**. Common rail **124** may be a straight pathway that extends along axis X through the center of the cooling chamber. In other embodiments, common rail **124** may be a curved pathway, an angled pathway, or another pathway which allows for pressured flow through common rail **124** and into cooling units **130** as described further herein.
(17) Spaced along common rail **124** are a plurality of cooling units **130**, so that each cooling unit **130** corresponds with an electrical component **115** as described further herein. Each cooling unit of the plurality of cooling units may be structurally similar. Descriptions of one cooling unit may apply to other cooling units in the plurality of cooling units. Cooling unit **130** comprises loop inlets **132**, cooling loops **134**, and loop outlets **150**. Cooling unit **130** may be sized and shaped to correspond with the size and shape of an electrical component **115** coupled to second side **114***a*. In some embodiments, there may be one cooling unit **130** arranged on first side **112***a* for each, minus one, electrical component **115** coupled to second side **114***a*, wherein the final electrical component **115** may be coupled to second side **114***a* opposite of collection cavity **160**. Similarly, there may be one cooling unit **130** arranged on first side **112***b* for each, minus one, electrical component **115** coupled to second side **114***b*, wherein the final electrical component **115** may be coupled to second side **114***b* opposite of collection cavity **160**. In other embodiments, there is one cooling unit **130** on first side **112***a* for each electrical component **115** coupled to second side **114***a*. Similarly, there may be one cooling unit **130** on a first side **112***b* for each electrical component **115** coupled to second side **112***b*. Each cooling unit **130** may be located directly opposite an electrical component **115** such that coolant within cooling unit **130** may absorb heat from the corresponding electrical component **115**.

(18) Common rail **124** may run through the center of each cooling unit **130**. Cooling unit **130** may include a plurality of loop inlets **132** and cooling loops **134**. Each cooling loop **134** corresponds with a loop inlet **132**. Coolant from common rail **124** flows through each of the loop inlets **132** and into the cooling loops **134**. Cooling unit **130** may be generally bisected by common rail **124**. Loop inlets **132** are fluidly connected and spaced apart on either side of common rail **124**. In one embodiment, as seen in FIG. **4**, each cooling unit **130** may include four cooling loops **134**. A first pair of cooling loops may be positioned on one side of common rail **124** and a second pair of cooling loops may be mirrored on the other side of common rail **124** opposite of the first two cooling loops. Loop inlets **132** may be positioned such that they are located on common rail **124** on first surface **112***a* opposite a middle region of corresponding electrical component **115** positioned on second surface **114***a*. Referring to the previous embodiment, four loop inlets **132** may be spaced apart along common rail **124** on first surface **112***a* opposite the middle region of electric component **115** such that a first pair of loop inlets **132** oppose a second pair of loop inlets **132** across common rail **124**.

(19) Each cooling loop **134** may be connected in parallel along common rail **124**. Cooling loop **134** begins at loop inlet **132** and ends at loop outlet **150**. A passageway **136** extends from loop inlet **132** to loop outlet **150** and is defined by passageway walls **137**. Coolant may flow down common rail **124** and enter each loop inlet **132** substantially simultaneously. Once coolant enters into cooling loop **134**, coolant then travels through passageway **136**. Passageway walls **137** may be sized and shaped to allow coolant to flow through passageway **136** such that a surface area of passageway **136** is completely immersed in coolant fluid. Passageway walls **137** may comprise a plurality of nubs **138** to facilitate turbulent flow of coolant through passageway **136** and increase surface area of passageway walls **137**. Similarly, a plurality of posts **170** may be spaced within passageway **136** spaced apart from passageway walls **137** to further facilitate turbulent flow of the coolant by interrupting flow. Turbulence in the coolant fluid may allow the coolant to absorb more heat from electronic components **115** by allowing even distribution of heat to the coolant.

(20) As shown in FIG. **3**, passageway walls **137** have a first height and posts **170** have a second height less than the first height. When first cooling device **110***a* and second cooling device **110***b* abut each other, as shown in FIG. **3**, the passageway walls **137***a* of first cooling plate **116***a* abut the passageway walls **137***b* of second cooling plate **116***b* to form enclosed passageway **136**, while posts **170***a* and **170***b* do not touch to further facilitate turbulence within passageway **136**.

(21) Passageway **136** may include a plurality of passageway segments **140** of various lengths. Each passageway segment **140** is coupled by a turn **139**. Passageway segments **140** may be positioned such that passageway **136** is serpentine and directs coolant back and forth across first surface **112***a*. In one embodiment, cooling loop **134** may include a passageway that comprises four serpentine passageway segments **140***a*, **140***b*, **140***c*, **140***d* coupled by three turns **139***a*, **139***b*, **139***c*. Turns **139** guide coolant flowing through each passageway segment **140** of passageway **136** back and forth on a portion of first surface **112***a* that corresponds to a portion of electronic component **115** coupled to second surface **114***a*. For example, each cooling loop **134** and its corresponding passageway **136** may, together with a portion of the common rail **124**, correspond generally with a quarter of corresponding electronic component **115**. The plurality of turns within passageway **136** facilitates even flow of the coolant within a compact corresponding section of the electronic component **115** and focuses flow of the coolant to allow for even heat distribution and efficiency.

(22) At the end of turns **139**, passageway **136** leads to loop outlet **150**. Loop outlet **150** is fluidly coupled to passageway **136** and collection cavity **160**. Coolant flows through cooling loop **134** via passageway **136** and exits loop outlet **150** into collection cavity **160**. Loop outlets **150** are sized and shaped such that a width of loop outlet **150** is smaller than a width of coolant inlet **122**. Due to the difference in width of the cooling inlet and the loop outlets, a pressure differential may be created, as discussed below.

(23) Once the coolant exits cooling loop **134** through loop outlet **150**, the coolant enters collection

cavity **160**. Collection cavity **160** may collect coolant that has run through cooling units **130**, which then exits collection cavity **160**. In some embodiments, an electronic component **115** may be positioned on second side **114** opposite collection cavity **160**, which is positioned on first side **112**. Coolant fluid may be drained from collection cavity **160** by primary output port or “cooling chamber outlet” **119**. Primary output port **119** may direct the coolant fluid used to cool electronic components **115** to other systems.

(24) In another embodiment, a common rail cooling device may comprise a first component **110***a* and a second component **110***b* configured to be coupled to first component **110***a* to form the common rail cooling device **100**. The one or more of first component **110***a* and second component **110***b* may include an outer surface **114** configured to receive one or more electronic components **115** to be cooled. Further, one or more of first component **110***a* and second component **110***b* may include wall section **137**, such that when the first and second components **110***a*, **110***b* are joined the wall sections **137** define within the common rail cooling device **100** a common rail cavity **124**, a collection cavity **160**, and a plurality of cooling units **130** adjacent to outer surface **114** configured to receive the one or more electronic components **115**, and wherein each cooling unit **130** includes a passageway **136** fluidly coupled between the common rail cavity **124** and the collection cavity **160**.

(25) A method for operating cooling device **110** is described herein. Cooling device **110** may be used to cool one or more electronic components **115** on second surface **114** of cooling device **110**. The method may include supplying coolant fluid to cooling chamber inlet **122**, and causing the coolant fluid to flow into common rail **124**, into and through the passageway **136** of each cooling unit **130**, out of each cooling unit **130** into collection cavity **160**, and out of cooling chamber **120** through cooling chamber outlet **119**.

(26) Referring to FIG. **5**, the pressure of the coolant in collection cavity **160** may be lower than the pressure of coolant flowing into cooling inlet **122**. Because of the pressure differential, coolant may be flowing at a much higher velocity through cooling inlet **122** into common rail **124** than coolant flowing into collection cavity from loop outlets **150**. Further, the difference in pressure between the high pressure cooling inlet **122** and low pressure collection cavity **160** may allow an even flow split of coolant into the plurality of loop inlets **132** located along common rail **124**. The high pressure coolant flowing in through cooling inlet **122** may cause the coolant to fill the common rail **124** quickly and evenly distribute coolant into loop inlets **134**. In other words, referring additionally to FIG. **6**, the high pressure within the common rail **124** causes the coolant to jet generally evenly into inlets **132** and corresponding cooling loops **134** at a high velocity. The even flow split of coolant may allow for each cooling unit **130** to receive the same amount of coolant to cool each corresponding electronic component **115**. Because each electrical component may correspond to a cooling unit **130** with an equal amount of coolant flowing through it, each electrical component is able to dissipate the same amount of heat. The size of collection cavity **160** and cooling inlet **122** may be scaled to support a variety of electronic components while maintaining the pressure differential.

(27) To better understand the description above, exemplary cooling assemblies **100***a* and **100***b* are described herein. As seen in FIGS. **2-4**, cooling assembly **100** comprises a first cooling device **110***a* and a second cooling device **110***b*. Cooling devices **110***a* and **110***b* are substantially alike. First cooling device **110***a* includes first plate **116***a* and cooling chamber **120***a*. Similarly, second cooling device **110***b* includes second plate **116***b* and cooling chamber **120***b*. Each cooling plate **116***a*, **116***b* has a first surface **112***a*, **112***b*, and a second surface **114***a*, **114***b*. Removably coupled to each second surface **114***a*, **114***b*, are a series of electronic components **115**, such as IGBT models. First surfaces **112***a* and **112***b* each comprise a cooling chamber **120***a*, **120***b*. Each cooling chamber is substantially alike. The first surface **112***a* of first cooling plate **116***a* and the first surface **112***b* of the second cooling plate **116***b* directly oppose each other such that cooling assemblies **120***a*, **120***b* are abutting. Referring to cooling device **110***a* in FIG. **4**, cooling chamber **120***a* comprises a cooling inlet **122**, a

common rail **124**, and three cooling units **130**. Each cooling unit **130** corresponds with one electrical component **115**. Each cooling unit **130** has four cooling loops **134** positioned such that a first pair of cooling loops are on one side of common rail **124** and a second pair of cooling loops are on the other side of common rail **124**. Each cooling loop **134** is fluidly connected to common rail **124** by a loop inlet **132**. Cooling inlets **132** are located on common rail **124** at a location on first surface **112***a* that is opposite the center of electronic component **115** coupled on second surface **114***a*. Each cooling loop, together with a portion of the common rail **124**, corresponds generally with one quadrant of electronic component **115**. High pressure coolant flows into common rail **124** from cooling inlet **122**. Coolant then evenly splits off into each one of loop inlets **132**. As coolant flows through loop inlets **132** into cooling loops **134**, the coolant passes through passageways **136**. Nubs **138** on passageway walls **137** and posts **170** within passageway **136** create turbulence in the coolant for even cooling as coolant flows through the three turns **139***a*, **139***b*, **139***c* of passageway **136**. At the end of passageway **136**, coolant exits cooling loop **132** through loop outlet **150** and enters the low-pressure collection cavity **160**. Each of the electronic components **115** are evenly cooled by the cooling chamber **120***a* on first surface **112***a*. Similarly, cooling device **110***b* has a cooling chamber **120***b* to cool the electronic components on second surface **114***b* of cooling plate **116***b*.

EXAMPLE

(28) FIG. **7** illustrates the thermal difference demonstrated between a conventional cooling device and cooling device **110** (FIGS. **2-4**) during a testing simulation. Cooling plate **200** corresponds with a first cooling plate **20** of cooling device **10**, wherein cooling plate **200** includes a plurality of electronic components **202**. Cooling plate **204** corresponds with a second cooling plate similar to cooling plate **20** of cooling device **10**, wherein cooling plate **204** includes a plurality of electronic components **206**. Cooling plate **208** corresponds with first cooling plate **116***a* of cooling device **110***a*, wherein cooling plate **208** includes a plurality of electronic components **210**. Cooling plate **212** corresponds with second cooling plate **110***b* of cooling device **110***b*, wherein cooling plate **212** includes a plurality of electronic components **214**.

(29) As shown, electronic component **202***a* of cooling plate **200** demonstrated an average temperature of 77.2° C., electronic component **202***b* demonstrated an average temperature of 80.3° C., and electronic component **202***c* demonstrated an average temperature of 79.7° C. The average temperature of electronic components **202** was 79.1° C.

(30) Electronic component **206***a* of cooling plate **204** demonstrated an average temperature of 77.3° C., electronic component **206***b* demonstrated an average temperature of 80.4° C., and electronic component **206***c* demonstrated an average temperature of 79.9° C. The average temperature of electronic components **206** was 79.2° C.

(31) Electronic component **210***a* of cooling plate **208** demonstrated an average temperature of 61.4° C., electronic component **210***b* demonstrated an average temperature of 61.4° C., and electronic component **210***c* demonstrated an average temperature of 62.1° C. The average temperature of electronic components **210** was 61.6° C.

(32) Electronic component **214***a* of cooling plate **212** demonstrated an average temperature of 61.4° C., electronic component **214***b* demonstrated an average temperature of 61.5° C., and electronic component **214***c* demonstrated an average temperature of 62.1° C. The average temperature of electronic components **214** was 61.7° C.

(33) The average temperature of the electronic components of the cooling assembly corresponding to conventional cooling assembly **10** was 79.1° C. The average temperature of the electronic components of the cooling assembly **100** was 61.7° C. As such, cooling assemblies **100***a* and **100***b* demonstrated a temperature difference of 17.4° C. compared to the conventional cooling assembly.

(34) While this invention has been described as having an exemplary design, the present invention may be further modified within the spirit and scope of this disclosure. This application is therefore intended to cover any variations, uses, or adaptations of the invention using its general principles.

Further, this application is intended to cover such departures from the present disclosure as come within known or customary practices in the art to which this invention pertains.

Claims

1. A cooling device, comprising: a surface configured to receive a component to be cooled; a cooling chamber inlet; a common central rail fluidly coupled to the cooling chamber inlet; a collection cavity; a plurality of cooling units, fluidly coupled to the common central rail such that each of the plurality of cooling units extends and is fed from both opposite sides of the common central rail and the common central rail bisects each of the plurality of cooling units; wherein each cooling unit of the plurality of cooling units is located adjacent to the surface and comprises a passageway fluidly coupled to the common central rail and the collection cavity; and wherein the cooling device is configured for a coolant fluid to only flow sequentially from the common central rail, through the plurality of cooling units and to the collection cavity; a cooling chamber outlet fluidly coupled to the collection cavity.
2. The cooling device of claim 1, wherein the common central rail, the plurality of cooling units and the collection cavity are configured such that a pressure of the coolant fluid within the common central rail is higher than a pressure of the coolant fluid within the collection cavity, and the coolant fluid flows into each of the plurality of cooling units at an equal velocity.
3. The cooling device of claim 1, wherein the passageway is serpentine, and defined by adjacent linear segments connected by turns.
4. The cooling device of claim 1, wherein each cooling unit of the plurality of cooling units further comprises a plurality of posts spaced apart from one another along a length of the passageway.
5. A common rail cooling device comprising: a first cooling plate, the first cooling plate comprising a first surface, and a second surface configured to receive one or more first electronic components; a cooling chamber on the first surface of the first cooling plate, the cooling chamber configured to receive a coolant fluid and comprising: a common rail chamber; a collection cavity; and a plurality of cooling units including passageways fluidly coupled between the common rail chamber and the collection cavity; wherein the plurality of cooling units are fluidly coupled to the common rail chamber such that each of the plurality of cooling units extends and is fed from both opposite sides of the common rail chamber and the common rail chamber bisects each of the plurality of cooling units; and wherein the common rail cooling device is configured for the coolant fluid to only flow sequentially from the common rail chamber, through the plurality of cooling units and to the collection cavity.
6. The common rail cooling device of claim 5, further comprising a second cooling plate opposite the cooling chamber of the first cooling plate, the second cooling plate including a first surface of the second cooling plate facing the cooling chamber of the first cooling plate and a second surface of the second cooling plate configured to receive one or more second electronic components.
7. The common rail cooling device of claim 6, wherein the one or more second electronic components on the second cooling plate and the one or more first electronic components on the first cooling plate are located adjacent to the plurality of cooling units.
8. The common rail cooling device of claim 7, wherein the passageways of each cooling unit includes a plurality of segments, each of the plurality of segments coupled to the next segment in the plurality of segments by a turn.
9. The common rail cooling device of claim 5, wherein the passageways are serpentine passageways.
10. The common rail cooling device of claim 5, wherein the plurality of cooling units are connected in a parallel fluid circuit between the common rail chamber and the collection cavity.
11. The common rail cooling device of claim 5, wherein the common rail chamber, the plurality of cooling units and the collection cavity are configured such that a pressure of the coolant fluid

within the common rail chamber is higher than a pressure of the coolant fluid within the collection cavity, and the coolant fluid flows into each of the plurality of cooling units at an equal velocity.
12. A cooling assembly, comprising: a first member comprising: a first cooling plate, the first cooling plate including a first surface of the first cooling plate, and a second surface of the first cooling plate configured to receive one or more first electronic components to be cooled; and a first plurality of wall segments extending from the first surface of the first cooling plate, the first plurality of wall segments defining a first common rail cavity portion, a first collection cavity portion, and a first plurality of cooling unit passageways between the first common rail cavity portion and the first collection cavity portion; and a second member coupled to the first member, comprising: a second cooling plate, the second cooling plate including a first surface of the second cooling plate, and a second surface of the second cooling plate configured to receive one or more second electronic components to be cooled; and a second plurality of wall segments extending from the first surface of the second cooling plate, the second plurality of wall segments defining a second common rail cavity portion, a second collection cavity portion, and a second plurality of cooling unit passageways between the second common rail cavity portion and the second collection cavity portion; wherein when the first member and the second member are coupled, the first plurality of wall segments are coupled to the second plurality of wall segments such that the first common rail cavity portion and the second common rail cavity portion form a common central rail, the first collection cavity portion and the second collection cavity portion form a collection cavity, and the first plurality of cooling unit passageways and the second plurality of cooling unit passageways form a plurality of cooling units between the common central rail and the collection cavity; wherein the cooling assembly is configured for a coolant fluid to only flow sequentially from the common central rail, through the plurality of cooling units and to the collection cavity; and wherein the plurality of cooling units are fluidly coupled to the common central rail such that each of the plurality of cooling units extends and is fed from both opposite sides of the common central rail and the common central rail bisects each of the plurality of cooling units.
13. The cooling assembly of claim 12, wherein: the first member comprises: a first cooling chamber inlet portion coupled to the first common rail cavity portion; and a first plurality of loop outlet portions each coupled to the first plurality of cooling unit passageways and the first collection cavity portion; and the second member comprises: a second cooling chamber inlet portion coupled to the second common rail cavity portion; and a second plurality of loop outlet portions coupled to the second plurality of cooling unit passageways and the second collection cavity portion; wherein when the first member and the second member are coupled such that the first plurality of wall segments are joined to the second plurality of wall segments, the first cooling chamber inlet portion and the second cooling chamber inlet portion define a cooling chamber inlet and the first plurality of loop outlet portions and the second plurality of loop outlet portions define a plurality of loop outlets.
14. The cooling assembly of claim 12, wherein each of the first plurality of cooling unit passageways and each of the second plurality of cooling unit passageways are configured such that a pressure of the coolant fluid within the common central rail is higher than a pressure of the coolant fluid within the collection cavity, and the coolant fluid flows into each of the first plurality of cooling unit passageways and the second plurality of cooling unit passageways at an equal velocity.
15. The cooling assembly of claim 12, wherein each of the plurality of first cooling unit passageways is a serpentine passageway formed from the first plurality of wall segments connected by turns and each of the plurality of second cooling unit passageways is a serpentine passageway formed from the second plurality of wall segments connected by turns.
16. The cooling assembly of claim 12, wherein each of the first plurality of cooling unit passageways and each of the second plurality of cooling unit passageways comprise a plurality of posts spaced apart from one another along a length of each of the first plurality of cooling unit

passageways and each of the second plurality of cooling unit passageways.
17. The cooling assembly of claim 16, wherein when the first member and the second member are coupled, the plurality of posts of the first member is spaced apart from a corresponding one of the plurality of posts of the second member by a distance.
18. The cooling assembly of claim 12, wherein the one or more first electronic components to be cooled on the first cooling plate is located adjacent to the first plurality of cooling unit passageways, and the one or more second electronic components to be cooled on the second cooling plate is located adjacent to the second plurality of cooling unit passageways.
19. The cooling assembly of claim 12, wherein the first plurality of cooling unit passageways is connected in a parallel fluid circuit between the first common rail cavity portion and the first collection cavity portion, and the second plurality of cooling unit passageways is connected in a parallel fluid circuit between the second common rail cavity portion and the second collection cavity portion.